

Risk Scoring Methodology

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1. Purpose

This document defines the quantitative methodology used to compute portfolio risk scores. The risk score is a numerical value from 1 (lowest risk) to 100 (highest risk) that summarizes a portfolio's overall risk exposure relative to the bank's risk framework.

2. Risk Score Calculation

The risk score is computed as a weighted sum of asset class risk factors:

$$\text{Risk Score} = \sum(\text{weight}_i * \text{risk_factor}_i) \text{ for each asset class } i$$

Asset class risk factors (on a 1-100 scale):

- Cash & Equivalents: 5
- Government Bonds (short-term): 10
- Government Bonds (long-term): 20
- Investment-Grade Corporate Bonds: 25
- High-Yield Bonds: 45
- Canadian Equities (large-cap): 50
- U.S. Equities (large-cap): 55
- International Equities (developed): 60
- Emerging Market Equities: 75
- Sector/Thematic ETFs: 70
- Alternative Investments: 80
- Cryptocurrency: 95

Example: A portfolio with 60% Canadian Equities and 40% Government Bonds (short-term) would have a risk score of: $(0.60 * 50) + (0.40 * 10) = 34$.

3. Risk Score Ranges and Profile Mapping

Risk Score Range	Risk Profile	Maximum Equity Exposure
1 - 20	Conservative	20%
21 - 45	Moderate	60%
46 - 70	Aggressive	90%
71 - 100	Speculative (requires special approval)	100%

4. Compliance Thresholds

A portfolio is considered out of compliance when its computed risk score exceeds the maximum allowed score for the client's assigned risk profile:

- Conservative clients: risk score must be ≤ 20
- Moderate clients: risk score must be ≤ 45
- Aggressive clients: risk score must be ≤ 70

If a portfolio exceeds the maximum risk score, the system flags it in the daily compliance report and the advisor is

required to take corrective action within the timelines specified in the Rebalancing Policy.

5. Limitations

This risk scoring model is a simplified linear model and does not account for correlations between asset classes, tail risk, or liquidity risk. It is intended as a screening tool to support suitability assessments, not as a replacement for comprehensive portfolio risk analysis. The risk factors are reviewed and updated semi-annually by the Risk Management Committee.

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